

# Metro Ticket Generating System

## SYSTEM REQUIREMENT ANALYSIS PHASE

|               |                                |
|---------------|--------------------------------|
| Date          | 26 July 2026                   |
| Name          | T V SAI LAXMI                  |
| Project Name  | Metro Ticket Generating System |
| Maximum Marks | 4 Marks                        |

## 7. Solution Requirements

### 7.1 Purpose

This document consolidates the functional and non-functional requirements for the Smart Metro Ticket Booking System. Requirements are derived from the Problem Statement, Empathy Map, and the finalized MVP ideas from the Brainstorming and Idea Prioritization document, and are scoped to be delivered using core ServiceNow platform capabilities.

### 7.2 Functional Requirements

| ID    | Requirement                                                                                                                            | Priority |
|-------|----------------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-01 | The system shall provide a Service Catalog item to capture source station, destination station, travel date, and number of passengers. | P0       |
| FR-02 | The system shall automatically calculate the fare based on station distance/zone and passenger count using Flow Designer.              | P0       |
| FR-03 | The system shall generate a unique digital ticket with an embedded QR code upon successful booking.                                    | P0       |
| FR-04 | The system shall send instant booking confirmation and the digital ticket via SMS/Email/Push notification.                             | P0       |
| FR-05 | The system shall maintain a booking history/status record for each user through the Service Catalog request record.                    | P0       |
| FR-06 | The system shall integrate with a payment gateway (UPI/Card/Wallet) for instant online payment.                                        | P1       |
| FR-07 | The system shall allow commuters to cancel a booking and trigger a refund workflow.                                                    | P1       |
| FR-08 | The system shall provide an admin dashboard for metro staff to monitor bookings, revenue, and station-wise traffic.                    | P1       |
| FR-09 | The system shall support season/monthly pass booking with auto-renewal reminders.                                                      | P2       |
| FR-10 | The system shall provide a Virtual Agent/chatbot for booking assistance and FAQs.                                                      | P2       |
| FR-11 | The system shall provide a voice-assisted accessibility mode for elderly/differently-abled commuters.                                  | P2       |
| FR-12 | The system shall integrate with station gate hardware/IoT scanners for automatic QR validation.                                        | P2       |

### 7.3 Non-Functional Requirements

| Category      | Requirement                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------|
| Performance   | Fare calculation and ticket generation shall complete within 5 seconds of payment confirmation.                        |
| Availability  | The booking service shall be available 24x7, removing dependency on physical counter hours.                            |
| Usability     | The booking flow shall be completable in under a minute on both desktop and mobile Service Portal views.               |
| Accessibility | The interface shall support screen-reader and voice-assistance compatibility for elderly and differently-abled users.  |
| Security      | Payment data shall be handled via a PCI-compliant gateway; no card details shall be stored on the ServiceNow instance. |
| Reliability   | The system shall handle booking failures gracefully with clear error messages and no duplicate charges.                |
| Scalability   | The solution shall support concurrent booking spikes during peak hours without performance degradation.                |
| Auditability  | Every booking, payment, and cancellation event shall be logged and traceable through the request record.               |

### 7.4 Assumptions

- Commuters have access to a smartphone or desktop with internet connectivity to use the booking portal.
- Station distance/zone data and fare rules are available and maintained as reference data on the platform.
- A third-party payment gateway is available for integration via REST APIs.
- Gate hardware/IoT integration for QR scanning is a future-phase enhancement and not required for MVP.

### 7.5 Constraints

- The solution must be built primarily using core ServiceNow platform components (Service Catalog, Flow Designer, Notifications).
- Phase 1 delivery is limited to the P0 (Must Have) MVP scope; P1–P3 items are planned for later phases.
- The system must comply with applicable data protection and payment security regulations.